

Enterprise Identity & Zero Trust Security Platform

Recruiter Project Overview | Portfolio Lab

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Executive Summary

A portfolio-grade enterprise security project demonstrating identity architecture, privileged access, PKI, certificate lifecycle management, workload identity, secure API access, cloud monitoring, infrastructure as code, testing, and governance.

Business Problem

A fictional enterprise requires secure access for employees, administrators, applications, and machines across cloud services. The solution applies Zero Trust principles: verify explicitly, use least privilege, and assume breach.

Solution Components

- Microsoft Entra ID identity, group, and application-role design
- Conditional Access and MFA policy catalogue with report-only rollout
- Private root and issuing CA architecture with certificate lifecycle automation
- OAuth/OIDC and mutual-TLS protected FastAPI service design
- Azure Key Vault and managed identity access pattern
- Terraform modules for network, vault, and monitoring foundations
- Microsoft Sentinel KQL analytics and incident response documentation
- Threat model, test cases, risk register, operations, DR, and compliance mapping

Technology Stack

Microsoft Azure, Microsoft Entra ID, Azure Key Vault, Microsoft Sentinel, KQL, PKI, OpenSSL, mTLS, OAuth 2.0, OpenID Connect, Terraform, PowerShell, Python, FastAPI, Docker, GitHub Actions, IAM, and Zero Trust.

Recruiter Review Path

1. Read the GitHub README and recruiter overview.
2. Review the solution architecture and threat model.
3. Inspect Terraform, Python, PowerShell, PKI, and KQL assets.
4. Review the test strategy and publication checklist.
5. View redacted lab evidence when added by the project owner.

Evidence and Integrity Statement

This is a fictional, synthetic-data portfolio lab. It does not contain employer information, production credentials, private keys, or fabricated deployment claims. Design files and starter implementation are included; live Azure results must be added only after execution in a personal lab.